

Deck Number Check

Verification of figures proposed for a client deck,
with an internally consistent replacement set

Client

Iwatani Corporation — Metals Department → Stainless Steel Division

Subject

Bellows tube (precision-slit material) market entry,
leveraging the Jindal Stainless relationship

Status

Working document — updated at monthly internal meetings

Prepared

July 2026

Evidence standard. Every factual claim in this document carries an inline source link and a verification grade: **[P]** primary (company, association or government publication), **[C]** credible press, **[S]** commercial market-research vendor (indicative only), **[U]** unverified directory listing. Commercial estimates for the bellows market disagree with one another by up to a factor of 19; figures marked **[S]** must not be quoted externally without stating the range. One proposed figure does not reconcile and should not be used as a point estimate.

13 — Deck Number Check

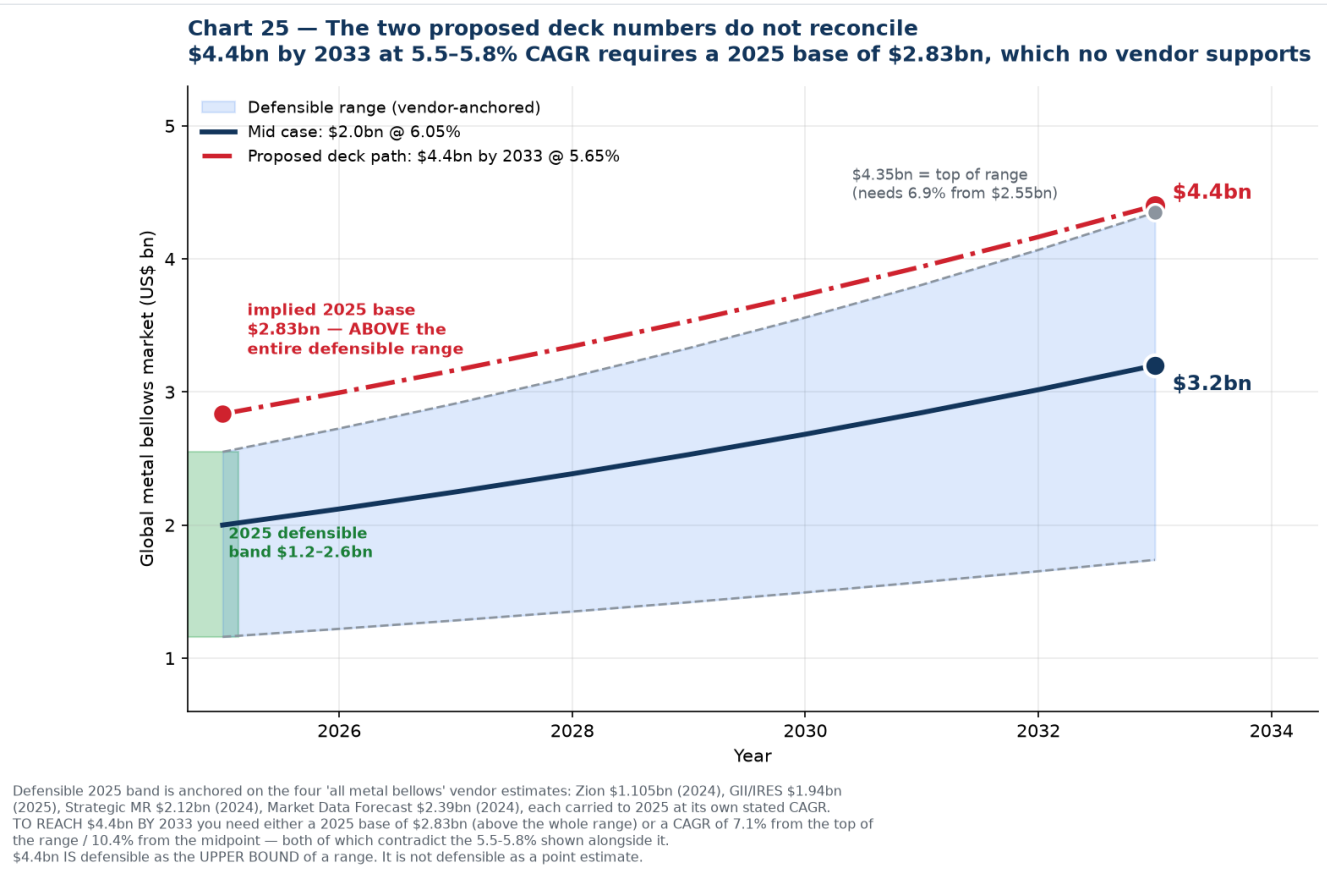
Verification of four figures proposed for a client deck.

Verdict at a glance

Proposed figure	Verdict
Global metal bellows, 2025 — (left blank)	Use US\$1.2-2.6 bn , midpoint ~ US\$2.0 bn
Forecast 2033: ~ US\$4.4 bn	Do not use as a point estimate. Defensible only as the top of a range
CAGR 2025-33: ~ 5.5-5.8%	Confirmed. Well supported — this is the modal cluster
India global share: ~ 2.7%	Confirmed as an estimate. Must be labelled modelled; no published figure exists

The core problem: the two numbers you have contradict each other.

1. Why US\$4.4 bn by 2033 fails



If 2033 is US\$4.4 bn and the CAGR is 5.5-5.8%, then working backwards the **2025 base must be US\$2.80-2.87 bn.**

That is above every vendor estimate for the global metal bellows market:

Vendor	Base-year value	Carried to 2025 at its own CAGR
Zion Market Research	US\$1.105 bn (2024)	US\$1.16 bn
GII / IRES	US\$1.94 bn (2025)	US\$1.94 bn
Strategic Market Research	US\$2.12 bn (2024)	US\$2.27 bn
Market Data Forecast	US\$2.39 bn (2024)	US\$2.53 bn
Defensible 2025 band		US\$1.16 - 2.55 bn

US\$2.83 bn sits **11% above the top of the range and 42% above the midpoint.**

Put the other way: what CAGR would you need to reach US\$4.4 bn?

Starting from 2025 base of...	Required CAGR to 2033
US\$1.16 bn (low)	18.1%
US\$2.00 bn (mid)	10.4%
US\$2.39 bn	7.9%
US\$2.55 bn (high)	7.1%

Every one of those exceeds the 5.5–5.8% written beside it. **You cannot hold both numbers at once.**

What US\$4.4 bn is defensible as

The top of the range. From the highest defensible 2025 base (US\$2.55 bn) at the highest vendor CAGR (6.90%), 2033 comes out at **US\$4.35 bn ≈ US\$4.4 bn**. So this phrasing works:

"The global metal bellows market is estimated at US\$1.2–2.6 bn in 2025, growing at 5–7% a year to somewhere between **US\$1.7 bn and US\$4.4 bn by 2033**."

What does not work is "the market will reach US\$4.4 bn by 2033," because that presents the most optimistic corner of a wide band as a central case, and it can be dismantled by anyone who divides one number by the other.

2. CAGR 5.5–5.8% — confirmed

This is the strongest of the four figures. It is not just inside the consensus band, it is the **modal cluster** of published estimates:

Vendor	CAGR
Zion Market Research	5.19%
Market Research Future	5.60%
Market Data Forecast	5.74%
GII / IRES	5.79%
Expert Market Research	5.90%
Dataintelo (welded)	6.20%
Dataintelo (hydroformed)	6.80%
Strategic MR / SkyQuest	6.90%

Five of eight land between 5.19% and 5.90%. **5.5–5.8% is well chosen.** If anything it is slightly conservative; ~6% is the true midpoint. Either is defensible.

3. India share ~2.7% — confirmed, with a labelling requirement

Correct as an estimate, and it is our figure. Two conditions on using it:

- 1. **Label it as an estimate.** No research vendor publishes an India bellows market share. Phrase it as "estimated at approximately 2–3%" rather than stating 2.7% as fact.
- 2. **It only reconciles with a ~US\$2.0 bn global base.** Cross-check: 2.7% of US\$2.0 bn = **US\$54 m**, which sits inside our independently derived India estimate of US\$50–70 m. But 2.7% of US\$2.83 bn (the base implied by your US\$4.4 bn figure) = **US\$76 m**, which falls outside that range. The inflated global number breaks the India number downstream too.

This is worth noting because it shows the error compounds: one over-stated figure makes two other slides inconsistent.

4. If you need a single citable source with a name on it

For a deck, a named source often beats a modelled range. The highest **sourced** 2033 figure for the all-metal-bellows scope is:

Market Data Forecast: US\$2.39 bn (2024) → US\$3.95 bn (2033), implied CAGR 5.74%

That gives you a 2033 number close to US\$4 bn, a CAGR inside your 5.5–5.8% target, and an attributable source. If the deck needs one line with a citation, use this and name the firm — adding that other estimates range from US\$1.1 bn to US\$2.4 bn for the base year because scope definitions differ.

5. Deck-ready numbers

DECK-READY NUMBERS — internally consistent, each traceable to a source			
Metal bellows market · use these exact figures and phrasings			
Global market size, 2025	US\$1.2 - 2.6 bn	midpoint ~US\$2.0 bn	Range of 4 vendors; scope definitions differ
Global forecast, 2033	~US\$3.2 bn	range US\$1.7 - 4.4 bn	Modelled from the 2025 band at consensus CAGR
CAGR, 2025 - 2033	~6%	range 5.2 - 6.9%	Every vendor falls inside this band
Highest SOURCED 2033 figure	US\$3.95 bn	Market Data Forecast	US\$2.39bn (2024) at 5.74% — cite by name
India market size, 2025	US\$50 - 70 m	≈ ₹435 - 610 cr	Bottom-up from MCA filings + 3 vendors
India share of global	~2 - 3%	point estimate 2.7%	MODELLED — no published figure exists
India growth, observed	+6.5%	FY24 → FY25, revenue-weighted	Measured from company filings — strongest number
DO NOT PUT "US\$4.4bn by 2033" ON A SLIDE AS A POINT ESTIMATE. It implies a 2025 base of US\$2.83bn, above every vendor estimate, and contradicts the 5.5-5.8% CAGR beside it. Use "~US\$3.2bn" as the figure, or "up to US\$4.4bn" as a range top.			

Metric	Use this	Range	Basis
Global market size, 2025	US\$1.2 - 2.6 bn	midpoint ~US\$2.0 bn	Four vendors; scope definitions differ
Global forecast, 2033	~US\$3.2 bn	US\$1.7 - 4.4 bn	Modelled from the 2025 band at consensus CAGR
CAGR, 2025-33	~6%	5.2 - 6.9%	Every vendor falls inside this band
Highest sourced 2033 figure	US\$3.95 bn	—	Market Data Forecast, cite by name
India market size, 2025	US\$50 - 70 m	≈ ₹435 - 610 cr	Bottom-up from MCA filings + three vendors
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India growth, observed	+6.5%	FY24 → FY25, revenue-weighted	Measured from company filings

Suggested slide wording

Global metal bellows market US\$1.2-2.6 bn (2025) · growing ~6% p.a. · ~US\$3.2 bn by 2033
Estimates vary by scope; range reflects four independent vendors

India US\$50-70 m (2025) · ~2-3% of global · +6.5% observed FY24→FY25 India share is our estimate; no published figure exists

Both lines survive scrutiny, and the caveats are short enough to sit in a footnote.

6. Two other figures to keep off the deck

Carried forward from earlier corrections in this pack, in case they are still circulating:

Figure	Status
"India holds 5-10% of the global expansion joints market"	Withdrawn. Implies a global market of US\$0.7-1.4 bn against four vendors at US\$3.7-7.2 bn. See 11
"India bellows CAGR ~6.9% (6Wresearch)"	Withdrawn. 6Wresearch publishes no market CAGR. See 10 §10
"North American semiconductor bellows market US\$1.2 bn"	Never use. Exceeds half the global all-industry market
"48% semiconductor + 56% aerospace of US demand"	Never use. Sums to 104%

7. The check to run on any future number

Before a figure goes on a slide, do the one-line reconciliation: **divide the forecast by the base and see whether the implied CAGR matches the CAGR you are quoting.** If they disagree, one of the three numbers is wrong.

That single division would have caught the US\$4.4 bn figure, the 5-10% expansion joints claim and the 104% end-use split. It takes ten seconds and it is the difference between a number that holds up in a partner meeting and one that does not.

Chart index

Chart	File	Purpose
25	25-deck-number-check.png	Why US\$4.4 bn and 5.5–5.8% cannot both hold
26	26-deck-card.png	Deck-ready figures, internally consistent

Generated by [charts/make_deck_card.py](#).